Shri G.S. Institute of Technology and Science

Department of Applied Mathematics and Computational Science

B.Tech I Year (July-Dec.2025) Group-B

Assignment II: MA 10509 (Mathematics for Data Science)

Last Date of Submission: 19/11/2025

	Questions	**CO**
Q.1	(i) Calculate Karl Pearson's coefficient of correlation from the following data:	**2**

X	6	8	12	15	18	20	24	28	31
Y	10	12	15	15	18	25	22	26	28

(ii) Calculate coefficient of correlation from the following data

x	12	9	8	10	11	13	7
y	14	8	6	9	11	12	3

Q.2	(i) The following are the ranks obtained by 10 students in commerce and accountancy are given below	**2**

Commerce	6	4	3	1	2	7	9	8	10	5
Accountancy	4	1	6	7	5	8	10	9	3	2

To what extent is the knowledge of students in the two subjects related?

(ii) A random sample of recent repair jobs was selected and estimated cost and actual cost were recorded.

Estimated cost	300	450	800	250	500	975	475	400
Actual cost	273	486	734	297	631	872	396	457

Calculate the value of spearman's correlation coefficient?

Q.3	**(i)** Calculate the two regression equations of X on Y and Y on X from the data given below, taking deviations from an actual means of X and Y.	**2**

Price(Rs.)	10	12	13	12	16	15
Amount Demanded	40	38	43	45	37	43

Estimate the likely demand when the price is Rs.20.

(ii) Find the means of X and Y variables and the coefficient of correlation between them from the following two regression equations: $4X–5Y+33 = 0$, $20X–9Y–107 = 0$.

Q.4	(i) From the data given below	**2**

Marks in Economics	25	28	35	32	31	36	29	38	34	32
Marks in Statistics	43	46	49	41	36	32	31	30	33	39

Find (a) The two regression equations, (b) The coefficient of correlation between marks in Economics and statistics, (c) The mostly likely marks in Statistics when the marks in Economics are 30.

(ii) In two independent samples of sizes 8 and 10 the sum of squares of deviations of the sample values from the respective sample means were 84.4 and 102.6. Test whether the difference of variances of the populations is significant or not. (The table value of F at 5%

	level for (7, 9) d.f. is 3.29).	
Q.5	(i) Define Random Sample with replacement and without replacement. A machine is designed to produce insulating washers for electric devices of average thickness of 0.025cm. A random sample of 10 washers was found to have an average thickness of 0.024cm with a standard deviation of 0.002cm. Test the significance of deviation using t-test. (Table value at 5% level is 2.262). (ii) A sample of 20 items has mean 42 units and standard deviation 5 units. Test the hypothesis that it is a random sample from a normal population with mean 45 units. (The tabulated value of t at 5% level for 19 d.f. is 2.615).	**2**
Q.6	(a) (i) A random variable X has the following probability function <table><tr><td>Values of X</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr><tr><td>Probability P(X)</td><td>a</td><td>3a</td><td>5a</td><td>7a</td><td>9a</td><td>11a</td><td>13a</td><td>15a</td><td>17a</td></tr></table>(i) Determine the value of 'a' (ii) Find P(X<3), P(X≥3), P(0<X<5) (iii) Find the distribution function of X. (b) When die is thrown, 'X' denotes the number that turns up. Find E(X), E(X^2) and Var (X).	**3**
Q.7	(i) For the following pdf, find c and mean of random variable X: $$f(x) = \begin{cases} c(x - x^2); 0 < x < 1 \\ 0; otherwise \end{cases}$$ (ii)A probability curve $y = f(x)$ has a range from 0 to ∞. If $f(x) = e^{-x}$, find the mean and variance.	**3**
Q.8	(i) A discrete Random variable X has the following probability mass function. <table><tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>P(X)</td><td>0.1</td><td>0.3</td><td>0.4</td><td>0.2</td></tr></table>Find mean E(X) and variance(X). (ii) The probability density function of X is $f(x) = \begin{Bmatrix} 3x^2, 0 \le x \le 1 \\ 0 \quad , otherwise \end{Bmatrix}$ Find mean E(X) and variance(X).	**3**
Q.9	(i) Define Random variable and its type, Probability density function and cumulative density function with examples. (ii) Define Hypothesis testing and Type-I and Type-II error with suitable example.	**3**
Q.10	Suppose X can take values 1, 2, 3, 4 with probabilities 0.1, 0.3, 0.4, 0.2 respectively. Write the **probability mass function** of X and verify that it is a valid PMF.	**3**